Random Subspace-Based k-Nearest Class Collaborative Representation for Hyperspectral Image Classification

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***Abstract*— Recently, collaborative representation classification (CRC) has attracted extensive interest for hyperspectral images (HSIs) classification. However, for collaborative representation with Tikhonov (CRT), a testing sample is collaboratively represented by training samples from all the classes, which may result in high computational cost. In this article, we select the first k class training samples that are nearest to the testing sample for representation, namely, k-nearest class CRT (KNCCRT) algorithm. In order to improve the performance of KNCCRT for HSI classification, the idea of random subspace-based KNCCRT ensemble framework is proposed. KNCCRT is adopted as base classifier and random subspace (RS) contributes to diversity by selecting feature randomly. Moreover, to further increase the classification accuracy, shape-adaptive (SA) neighborhood constraint is utilized in RS ensemble framework to incorporate spatial information. Experimental results on three real hyperspectral data sets demonstrate the effectiveness of the proposed methods for HSI classification. The combination of KNCCRT and RS framework provides a reliable accuracy for HSI classification.**

***Index Terms*— Collaborative representation, ensemble learning, hyperspectral data, k-nearest class, random subspace (RS), shape-adaptive (SA) neighborhood.**

I. Introduction

HYPERSPECTRAL image (HSI) sensors obtain a wide range of contiguous narrow spectral bands with very fine spectral resolution. They can provide rich spectral information, which is widely applied in recognition, target detection, and classification. Classification is one of the critical research topics in HSI analysis. However, the fundamental challenges in HSI classification task include insufficient training samples and high dimensionality [1]–[3]. In order to tackle these challenges, numerous advanced methods including representation-based model, ensemble learning framework, and spatial–spectral fusion strategy have been developed over the past few years.

Representation-based methods have been a research hot topic for HSI classification in recent years [4], [5]. Its main idea is that training samples can linearly represent testing samples. Sparse representation classification (SRC) is initially proposed for face recognition, and then successfully applied in HSI classification. In SRC model, a testing sample is approximated as a sparse linear combination of training samples by adopting l_1-norm regularization [6]. But recent studies have proved that it is not the sparsity constraint but the collaborative nature that actually enhances classification performance [7]. For collaborative representation classification (CRC), all training samples collaboratively represent a testing sample via l_2-norm regularization. Compared with SRC, CRC provides much lower computation since it obtains a closed-from solution [8]. There are also many representation-based classifiers for HSI analysis, for instance, the nearest regularized subspace (NRS) classifier [9], sparse representation-based nearest neighbor (SRNN) classifier [10], kernel collaborative representation with Tikhonov regularization (KCRT) classifier [11], and weighted-fusion-based representation classifier [12]. In particular, by introducing distance-based Tikhonov regularization term, collaborative representation with Tikhonov (CRT) has achieved excellent performance for HSI classification. However, representation-based method is greatly influenced by the selected dictionary.

For ensemble learning, multiple base learners are integrated into a strong learner via an appropriate strategy to achieve better classification performance and generalization than any base learner [13]–[16]. It requires the classification accuracy of the base learners to be higher than 50% in ensemble learning. The two vital components are accuracy and diversity in ensemble system [17]. In order to obtain diversity, different classifiers are selected for heterogeneous ensemble. But for homogeneous ensemble, it is usually to manipulate the training data and input features. Among them, the representative ensemble methods are Bagging, Boosting, and random subspace (RS) [18]–[20]. For Bagging and Boosting, they are based on training samples [21], [13]. RS is one of the most popular ensemble methods, in which each base learner is trained on different

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feature subsets by selecting the feature randomly without replacement [16]. A complex high dimensional problem can be decomposed into several simple lower dimensional problems. Thus, RS can reduce the redundancy between features and alleviate the curse of dimensionality [22], [23]. The most common RS ensemble algorithms for HSI classification are random forest (RF) [24], [25]. A random-selection-based support vector machine (SVM) is developed for HSI classification [26]. However, the random selection approach may result in instability. To overcome this issue, the clustering strategy is adopted in RS ensemble. Also, the genetic algorithm (GA) and Jeffries–Matusita (JM) distance criteria are combined with feature selection to optimize the SVM subspace [27]. Based on the sparse representation (SR) mechanism, an SR-based RS ensemble framework is explored [28]. The RS ensemble with enhanced feature (RSE-EF) method is proposed in [29], which trains several classifiers based on enhanced RS. A deep learning-based RS ensemble is designed [30], [31], which further develops the ensemble learning. A novel deep subspace clustering method is proposed in [32], which can learn the distribution-preserving latent representation, and enables strong capacity for subspace clustering. However, RS-based ensemble models often ignore the spatial information of HSI image which limits the classification accuracy ineluctably.

It is noted that the spectral–spatial technique can effectively improve classification reliability. Besides the spectral information, the spatial information derived from the neighboring pixels is also considered in the algorithm. The integration of spectral–spatial features can provide valuable complementary information [33]. For instance, collaborative representation incorporated neighborhood information (JCR) is proposed [34]. Considering the importance of each neighborhood pixel, a weight version of JCR, namely, WJCR is developed [35]. In this algorithm, it assigns appropriate weights for the neighboring pixels around the central pixel. Instead of selecting one single region scale, a multiscale JCR with locally adaptive dictionary (MLJCRC) is designed [36]. This method takes into account multiscale and locally adaptive dictionary simultaneously. To address the shortcoming caused by the fixed window in MLJCRC, SALJCRC is proposed to use shape-adaptive (SA) neighborhood to better reflect the complex spatial information [37]. Spatial–spectral technique often fails to explore the spatial information from training samples adaptively and effectively.

In CRT model, some classes that significantly different from the testing sample may produce negative impact. Thus, k-nearest class CRT (KNCCRT) algorithms is proposed to optimize the training dictionary. The optimal k value is difficult to determine in KNCCRT algorithm. In order to address this problem and improve classification accuracy, ensemble learning integrated with KNCCRT is explored in this article. Nevertheless, traditional RS-based ensemble algorithms rarely consider local spatial information. So, an RS-based ensemble model with spatial information constraints is further developed. Existing spatial–spectrum fusion algorithm only constructs the SA local region of the testing sample, ignoring the fact that the training samples also contain spatial information. Therefore, the SA neighborhood of training and testing samples is designed in this article, which can make full use of spatial information to offer better classification accuracy.

In this article, KNCCRT algorithm is proposed first for HSI classification. It utilizes nearest-classes' training samples to optimize training dictionary, which can enhance self-similarity in reconstruction and computing efficiency simultaneously. Then, RSKNCCRT is proposed by combining KNCCRT with RS ensemble framework. It can not only address the curse of dimensionality in HSI classification but also avoid the choice of the optimal value of k. By setting the range of k, k is selected randomly during ensemble process, which can enhance the diversity between base classifiers. Furthermore, SA region is selected to sufficiently capture the complex neighborhood spatial structure. So, the SA regions of training and testing sample are constructed adaptively. Then, average filtering is employed for HSI to estimate spatial consistency.

The main contributions of this article are as follows: 1) KNCCRT algorithms are designed, which use k-nearest neighborhood technique to optimize dictionary; 2) an RS ensemble-based KNCCRT framework is developed for addressing high dimensionality in HSI classification task; and 3) furthermore, an SA constraint can further explore the spatial structure information of testing and training sample.

The outline of this article is organized as follows. Section II introduces CRT, RS, and SA neighborhood. Section III proposes the KNCCRT, RS-based KNCCRT (RSKNCCRT) algorithms, and SA neighborhood-based RSKNCCRT algorithms (SARSKNCCRT). In Section IV, experiment and analysis are presented on three public hyperspectral datasets. Section V discusses the characteristics of the proposed algorithms and parameter sensitivity. Finally, Section VI draws the conclusion and discusses future research.

II. Background

A. CRT

All the training samples are used to construct a training dictionary in the original CR model [38]. It applies an l_2-norm regularization to constrain the collaborative coefficient. Training samples $\boldsymbol{X} = \{\boldsymbol{x}_m\}_{m=1}^{c} \in \mathbf{R}^{N\times M}$ represent the N-dimensional data sets, which consists of M training samples. Note that c refers to the total number of classes. The training dictionary $\boldsymbol{D} = [\mathrm{D}_1, \mathrm{D}_2, \ldots, \mathrm{D}_c]$ is constructed based on training samples $\boldsymbol{X}$. The corresponding subdictionary $\boldsymbol{D}_m$ ($m \in \{1, 2, \ldots, c\}$) is denoted by $\mathrm{D}_m = [\boldsymbol{x}_{m,1}, \boldsymbol{x}_{m,1}, \ldots, \boldsymbol{x}_{m,M_m}]$, and thus $\sum_{m=1}^{c} M_m = M$. Suppose that a testing sample is denoted by $\boldsymbol{y} \in \boldsymbol{R}^N$, the objective function of CRC can be written as

$$\boldsymbol{\alpha} = \arg\min_{\boldsymbol{\alpha}^*} \left(\|\boldsymbol{y} - \boldsymbol{D}\boldsymbol{\alpha}^*\|_2^2\right) + \lambda \left\|\boldsymbol{\alpha}^*\right\|_2^2. \tag{1}$$

The representation coefficient vector $\boldsymbol{\alpha} \in \boldsymbol{R}^M$ is obtained in a closed-form solution

$$\boldsymbol{\alpha} = (\boldsymbol{D}^T\boldsymbol{D} + \lambda \boldsymbol{I})^{-1}\boldsymbol{D}\boldsymbol{y} \tag{2}$$

where $\boldsymbol{I}$ denotes the identity matrix. After solving the representation coefficient vector $\boldsymbol{\alpha}$, the class label of the testing

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sample $\boldsymbol{y}$ is determined according to minimum reconstruction error criterion

$$\text{class}(\boldsymbol{y}) = \arg\min_{m=1,2,\ldots,c} \|\boldsymbol{y} - \boldsymbol{D}_m \alpha_m\|. \tag{3}$$

By adding the distance-based Tikhonov regularization term, the optimization problems of CRT can be rewritten as follows:

$$\boldsymbol{\alpha} = \arg\min_{\boldsymbol{\alpha}} \|\boldsymbol{y} - \boldsymbol{D}\boldsymbol{\alpha}\|_2^2 + \lambda \|\boldsymbol{\Gamma}\boldsymbol{\alpha}\|_2^2 \tag{4}$$

where λ is the global regularization parameters to tradeoff between the reconstruction residual part and locality prior. Here, the expression of $\boldsymbol{\Gamma}$ is formulated as (5), which obtained by computing the distance between training sample $\boldsymbol{x}$ and testing sample $\boldsymbol{y}$. If the distance between $\boldsymbol{x}$ and $\boldsymbol{y}$ is the smallest, then it will play more important role in representing the testing sample $\boldsymbol{y}$, and vice versa

$$\boldsymbol{\Gamma} = \begin{bmatrix} \|\boldsymbol{y} - \boldsymbol{x}_1\|_2 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \|\boldsymbol{y} - \boldsymbol{x}_M\|_2 \end{bmatrix}. \tag{5}$$

The representation coefficient vector $\boldsymbol{\alpha} \in \boldsymbol{R}^M$ can be solved with a closed-form solution as

$$\boldsymbol{\alpha} = (\boldsymbol{D}^T \boldsymbol{D} + \lambda \boldsymbol{\Gamma}^{\mathrm{T}} \boldsymbol{\Gamma})^{-1} \boldsymbol{D}^T \boldsymbol{y}. \tag{6}$$

After obtaining the representation coeffective vector $\boldsymbol{\alpha} = \{\boldsymbol{\alpha}_1, \boldsymbol{\alpha}_2, \ldots, \boldsymbol{\alpha}_c\}$ per class, the class label of $\boldsymbol{y}$ is assigned to the class that producing the minimum representation residual

$$\text{class}(\boldsymbol{y}) = \arg\min_{m=1,2,\ldots,c} \left(\|y - D_m \alpha_m\|_2^2\right). \tag{7}$$

B. Random Subspace

The RS ensemble is one of the most effective yet simple ensemble frameworks in multiple classifier systems (MCSs). In this method, feature subsets are selected randomly without replacement from the whole original high-dimensional feature space. Then, low-dimensional feature subsets are generated by instructive iteration and a base classifier is constructed based on each feature subset. Each base classifier assigns the label of a testing sample independently. Finally, the final class result is determined by using a majority voting rule. The pseudo-code of the RS algorithm is described as follows.

Input: Training samples $D = \{(\boldsymbol{x}_1, l_1), (\boldsymbol{x}_2, l_2), \ldots, (\boldsymbol{x}_m, l_m)\}, l_i \in \{1, 2, \ldots, c\}, l_i$ represents the label of training samples, c represents total classes

Base learner H, feature set: S, number of base classifiers T

For t = 1, 2, ... T

1.Randomly select bands from the original feature S without replacement

2.Generate a new feature subset S_t from the original feature S

3.Train a base learner from the new feature subset $h_t = H(\mathrm{S}_t)$

End

Output: $H(x) = \arg\max_{l \in \{1,2,\ldots,K\}} (\sum_{t=1}^{T} \mathrm{I}(h_t(y) = l))$, if $h_t(\boldsymbol{y}) = l$, $\mathrm{I}(h_t(\boldsymbol{y}) = l) = 1$, otherwise, the value is 0

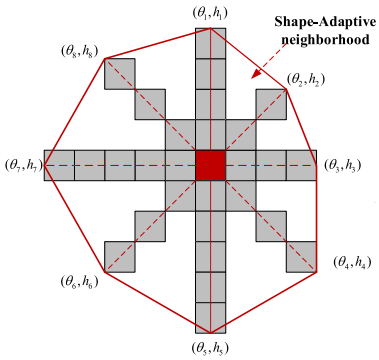


Fig. 1. Illustration of SA region.

C. Shape-Adaptive Neighborhood

Assuming the target sample is located at the center of neighborhood, it emits rays in eight directions and then connects the vertices of the rays to obtain a local SA region [39]. The detailed diagram is shown in Fig. 1.The eight directions θ_i(45° interval and$i = 1, 2, \ldots, 8$) are known and the eight lengths $h_i, i = 1, 2, \ldots, 8$ from target sample to vertices need to be calculated. The algorithm consists of two main steps. First, local polynomial approximation (LPA) filter kernels function $\{g_{h_k}, \theta_k\}$ is applied to the first principle component (PC1) of HSI. And, a set of smoothness density estimate and error is obtained in eight directions [38]

$$\left\{X_{h_i,\theta_i}^{\mathrm{PC1}}\right\} = X^{\mathrm{PC1}} \otimes \{g_{h_i}, \theta_i\} \tag{8}$$

$$\varepsilon_{h_i,\theta_i} = \left|X_{h_i,\theta_i}^{\mathrm{PC1}} - X_{\mathrm{noise_reduced}}^{\mathrm{PC1}}\right|. \tag{9}$$

Then, confidence space of ray length is computed, and the expression is formulated as (10). The intersection of confidence intervals (ICIs) rule is employed to determine the optimal ray length in eight directions. Finally, the local SA region of target sample is constructed by connecting endpoints of eight rays.

SA neighborhood is in a pointwise manner, in which a target sample corresponds to a unique local SA neighborhood. Therefore, the local SA neighborhood for each sample may be different, and it may often overlap

$$E_{h_i,\theta_i} = \left[\hat{X}_{h_i,\theta_i}^{\mathrm{PC1}} - \varepsilon_{h_i,\theta_i}, \hat{X}_{h_i,\theta_i}^{\mathrm{PC1}} + \varepsilon_{h_i,\theta_i}\right] \tag{10}$$

$$\tilde{h} = \{h_i | E_{h_i,\theta_i} \cap E_{h<h_i,\theta_i} \neq \emptyset \& E_{h_i,\theta_i} \cap E_{h_{I+1},\theta_i} = \emptyset, \forall h_i \in H\}. \tag{11}$$

III. Proposed Methods

A. KNCCRT

In CRT model, a testing sample is approximated by using the training samples from all of classes [40]. However, some classes that significantly different from the testing sample may produce negative effect. So, it is more rational to use a part of classes training samples to represent. Specifically, we use the first k class that is similar to the testing sample. In general, k is a pre-given constant and less than the total number of classes ($k < c$). Essentially, CRT can be considered as a special case

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of KNCCRT algorithm, when k is equal to the total number of classes. Irrelevant training classes are removed, so it can make CRT more simplified and less time-consuming simultaneously.

The KNCCRT algorithm attempts to obtain the k-nearest class by calculating the distance between testing sample $\boldsymbol{y}$ and subdictionary $\boldsymbol{D}_m (m = 1, 2, \ldots, c)$, $\boldsymbol{dis} = [\mathrm{dis}_1, \mathrm{dis}_2, \ldots, \mathrm{dis}_K]$. And dis_m is expressed as

$$\mathrm{dis}_m = \sum_{M_{m-1}}^{M_m} \|\boldsymbol{D}_m - \boldsymbol{y}\|_2^2. \tag{12}$$

The key of KNCCRT is to find the first k minimum values of dis by sorting $\boldsymbol{dis}$. The smaller the value of dis_m, the more similar between $\boldsymbol{y}$ and this subdictionary. Once the first k minimum values of dis are determined, that is, a subset of k training samples is chosen for constructing new dictionary. The train data sets turn to $\boldsymbol{X} \in \boldsymbol{R}^{N \times M_k}$. Then, corresponding training dictionary $\hat{\boldsymbol{D}}_k = [\boldsymbol{D}_1, \boldsymbol{D}_2, \ldots, \boldsymbol{D}_k]$ is built based on new train data sets. The optimization problem of KNCCRT is modified as follows:

$$\boldsymbol{\alpha} = \arg\min \|\boldsymbol{y} - \hat{\boldsymbol{D}}_k \boldsymbol{\alpha}\|_2^2 + \lambda \|\hat{\boldsymbol{\Gamma}}_{\boldsymbol{k}} \boldsymbol{\alpha}\|_2^2. \tag{13}$$

A diagonal weight matrix $\hat{\boldsymbol{\Gamma}}_k$ is added in (13). It adjusts the regularization parameter adaptively according to the similarity. $\hat{\boldsymbol{\Gamma}}_k$ is denoted by

$$\hat{\boldsymbol{\Gamma}}_k = \begin{bmatrix} \|\boldsymbol{y} - \boldsymbol{x}_1\|_2 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \|\boldsymbol{y} - \boldsymbol{x}_{M_k}\|_2 \end{bmatrix}. \tag{14}$$

The representative coefficient vector $\boldsymbol{\alpha}$ in (13) can be analytically derived as

$$\boldsymbol{\alpha} = \left(\hat{\boldsymbol{D}}_k^T \hat{\boldsymbol{D}}_k + \lambda \hat{\boldsymbol{\Gamma}}_{\boldsymbol{k}}^T \hat{\boldsymbol{\Gamma}}_{\boldsymbol{k}}\right)^{-1} \hat{\boldsymbol{D}}_k \boldsymbol{y}. \tag{15}$$

After obtaining the representation coefficient per class $\boldsymbol{\alpha} = [\boldsymbol{\alpha}_1, \boldsymbol{\alpha}_2, \ldots, \boldsymbol{\alpha}_k]$, the approximation reconstructed from each subdictionary is calculated subsequently. Then, the testing sample $\boldsymbol{y}$ can be classified to the class that producing the minimal residual between approximation and testing sample

$$\mathrm{class}(\boldsymbol{y}) = \arg\min_{m=1,2,\ldots,k} (\|y - D_m \alpha_m\|_2^2). \tag{16}$$

The main steps of KNCCRT are summarized in Algorithm 1.

Algorithm 1 KNCCRT

Input: $D = [D_1, D_2, D_3, \ldots, D_c]$: training set, λ: regularized parameter, $\boldsymbol{y}$: testing sample, k: the number of predefined class

1.Compute the distance between $\boldsymbol{y}$ and training samples per class

2.Sort $\boldsymbol{dis}$ to find the k-nearest classes of $\boldsymbol{y}$ to obtain $\hat{\boldsymbol{D}}_k$

3.Solve $\boldsymbol{\alpha} = (\hat{\boldsymbol{D}}_k^T \hat{\boldsymbol{D}}_k + \lambda \hat{\boldsymbol{\Gamma}}_k^T \hat{\boldsymbol{\Gamma}}_k)^{-1} \hat{\boldsymbol{D}}_k \boldsymbol{y}$

4.Compute the residual $\boldsymbol{r}_m = \|\boldsymbol{y} - \boldsymbol{D}_m \boldsymbol{\alpha}_m\|_2^2, \boldsymbol{m} = 1, \ldots, \boldsymbol{k}$

Output: Assign the class membership to $\boldsymbol{y}$ according to class($\boldsymbol{y}$)=arg min$\{\boldsymbol{r}_m\}$

B. RSKNCCRT

RS is also named random feature selection (RFS) and attribute bagging. RS aims at generating several feature subsets by sampling the original feature set randomly. The subspace is composed by randomly selected feature subset in each iteration. Each base classifier is trained on a subspace and assigns the class label of the testing sample, respectively. Then, diverse classification results are combined to create the final class results by a majority voting rule.

In the RS-based KNCCRT ensemble, the feature subset is generated by selecting the bands from the original feature space $\boldsymbol{X} \in \boldsymbol{R}^{N \times M}$ without replacement. Then, feature subset turns to $\tilde{\boldsymbol{X}} = [\tilde{\boldsymbol{x}}_1, \tilde{\boldsymbol{x}}_2, \ldots, \tilde{\boldsymbol{x}}_M]$, where $\tilde{\boldsymbol{x}}_i \in \boldsymbol{R}^{N'} (N' = d \times N)$. Obviously, the size of feature subset decrease, which can reduce the correlation between features effectively.

In the ensemble procedure, a new feature subset $\tilde{\boldsymbol{X}}_t \in \boldsymbol{R}^{N' \times M}$ is used to build a training dictionary $D_t = \{D_{1t}, D_{2t}, D_{3t}, \ldots, D_{ct}\}$. Then, KNCCRT algorithm is applied in each subspace. Specifically, $\boldsymbol{dis}_m$ is computed and then ranked. After obtaining the k-nearest class of $\boldsymbol{y}$, the new training dictionary $\hat{\boldsymbol{D}}_{kt} = [\boldsymbol{D}_{1t}, \boldsymbol{D}_{2t}, \ldots, \boldsymbol{D}_{kt}]$ is constructed based on a new training sets $\boldsymbol{X}_t \in \boldsymbol{R}^{N' \times M_k}$. Due to sampling randomly, different subspaces are obtained for each iteration. So, in each subspace, the optimum representation of the testing sample may be from different classes. Thus, the label of the testing sample may be different in each subspace. Each KNCCRT classifier predicts the label of the testing sample independently. After iterating T times, the set of labels of testing sample $\boldsymbol{l} = [l_1, l_2, l_3, \ldots, l_T]$ is obtained. Finally, the final class label of the testing sample is obtained according to a majority voting rule. The main steps of RSKNCCRT are illustrated in Algorithm 2.

In the KNCCRT algorithm, the selection of the optimal k is a troublesome problem. But in RSKNCCRT model, we can

Algorithm 2 RSKNCCRT

Input: KNCCRT: base classifier, T: the number of classifiers, $\boldsymbol{X} \in \boldsymbol{R}^{N \times M}$: input training sets, λ: regularized parameter, $\boldsymbol{y}$: testing sample, d: the number of features in a subspace ($N' = d \times N$), $\boldsymbol{S}$: feature set, k: pre-defined class number

For t=1 to T

1.Randomly selected from feature set **S**without replacement to form a new training set $\boldsymbol{X}_t \in \boldsymbol{R}^{N' \times M}$ composed of N' features

2.Compute the $\boldsymbol{dis}_m = \min\{\|\boldsymbol{D}_m - \boldsymbol{y}\|_2\}, m = 1, \ldots, c$

3. Sort $\boldsymbol{dis}_m$ to find the k-nearest class of $\boldsymbol{y}$ to obtain $\hat{\boldsymbol{D}}_{kt}$

4.Construct a new dictionary

$\hat{D}_{kt} = [D_{1t}, D_{2t}, D_{3t}, \ldots, D_{kt}]$ by new training set $\boldsymbol{X}_t \in \boldsymbol{R}^{N' \times M_k}$

5.Assign the class membership l_t to $\boldsymbol{y}$

End for

Return label matrix: $\boldsymbol{l} = [l_1, l_2, l_3, \ldots, l_T]$

Determine the final membership using the majority voting rule

Output: class($\boldsymbol{y}$) = mode ($\boldsymbol{l}$)

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set the range of k, and k is selected randomly during ensemble process, which can increase the diversity between base classifiers and avoid the selection of the optimal k simultaneously.

C. SARSKNCCRT

In HSI analysis, adjacent pixels have high spectral similarity and are likely to belong to the same class [41], [3]. To utilize such local homogeneity, it is straightforward to average all the pixels with its surrounding neighbor pixels. However, a fixed neighborhood cannot sufficiently capture the complex spatial structure in an HSI image. In fact, samples in homogeneous areas require large regularly shaped neighborhood, while samples in heterogeneous areas may require small irregularly shaped neighborhood. Thus, the SA neighborhood can be applied to faithfully reflect the complex spatial structure in an HSI image, especially for edges and details. As described in Section II, spatial information can be exploited by designing an SA neighborhood for training and testing sample in this article. Then, the average filtering is applied for the whole HSI image to estimate spatial consistency.

The testing sample $\boldsymbol{y}$ is located at the center of an SA region with n_{SA} neighbors $\bar{\boldsymbol{y}}_{\mathrm{SA}} = (1/n_{\mathrm{SA}}) \sum_{i=1}^{n_{\mathrm{SA}}} \boldsymbol{y}_i$, where $\bar{\boldsymbol{y}}_{\mathrm{SA}}$ represents the average spectral value for the central sample. The training samples $\boldsymbol{x}$ is located at the center of an SA neighborhood with n_{SA} neighbors, $\bar{\boldsymbol{x}}_{\mathrm{SA}} = (1/n_{\mathrm{SA}}) \sum_{\mathrm{i}=1}^{n_{\mathrm{SA}}} \boldsymbol{x}_i$, where $\bar{\boldsymbol{x}}_{\mathrm{SA}}$ represents the average spectral value for the central sample. The average of surrounding samples in a local SA region replaces the center sample to sufficiently utilize the spatial information. SARSKNCCRT is illustrated in Fig. 2

$$\bar{\boldsymbol{x}}_{\mathrm{SA}} = (1/n_{\mathrm{SA}}) \sum_{\mathrm{i}=1}^{n_{\mathrm{SA}}} \boldsymbol{x}_i \tag{17}$$

$$\bar{\boldsymbol{y}}_{\mathrm{SA}} = (1/n_{\mathrm{SA}}) \sum_{i=1}^{n_{\mathrm{SA}}} \boldsymbol{y}_i. \tag{18}$$

The objective function of SARSKNCCRT is expressed as

$$\boldsymbol{\alpha} = \arg\min \left\| \bar{\boldsymbol{y}}_{\mathrm{SA}} - \hat{\boldsymbol{D}}_k \boldsymbol{\alpha} \right\|_2^2 + \lambda \|\boldsymbol{\Gamma}_{\boldsymbol{k}} \boldsymbol{\alpha}\|_2^2. \tag{19}$$

The distance-based regularization term $\boldsymbol{\Gamma}_{\boldsymbol{k}}$ is denoted in the form of

$$\boldsymbol{\Gamma}_{\boldsymbol{k}} = \begin{bmatrix} \left\| \bar{\boldsymbol{y}}_{\mathrm{SA}} - \bar{\boldsymbol{x}}_{\mathrm{SA1}} \right\|_2 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \left\| \bar{\boldsymbol{y}}_{\mathrm{SA}} - \bar{\boldsymbol{x}}_{\mathrm{SAM}_k} \right\|_2 \end{bmatrix}. \tag{20}$$

Compared with the RSKNCCRT algorithm, the training sample $\boldsymbol{x}$ and testing sample $\boldsymbol{y}$ are replaced with $\bar{\boldsymbol{x}}_{\mathrm{SA}}$ and $\bar{\boldsymbol{x}}_{\mathrm{SA}}$ in SARSKNCCRT, respectively. The other steps of both methods are consistent

IV. Experiments and Results

A. Experimental Data

To evaluate the performance of the proposed models, three real hyperspectral data sets are considered for classification in the experiments. The first hyperspectral data set, Indian Pines, is acquired by the AVIRIS sensor over the Indian Pines region in June 1992. The rural scene contains 145 × 145 pixels with moderate spatial resolution of 20 m. It provides 200 spectral reflectance bands after removing 20 water absorption bands in the spectral range of 0.4–2.5 μm. Nine land cover classes are selected in the experiments. The false color composite image and the ground reference map are shown in Fig. 3(a) and (b), respectively.

TABLE I
Parameters Setting

Data		Indian Pines	Pavia University	Salinas
KNCCRT	λ	1e-1	1e-1	1e-2
	k	8	8	8
RSKNC CRT	λ	1e-3	1e-2	1e-4
	d	0.2	0.2	0.1
	k	7,8,9	7,8,9	8,9,10
	T	20	20	20
SARSKNC CRT	λ	1e-5	1e-4	1e-4
	d	0.2	0.2	0.2
	k	7,8,9	7,8,9	8,9,10
	T	20	20	20

The second hyperspectral data set, Pavia University, is acquired by Reflective Optics Spectrographic Imaging System (ROSIS) sensor in the University of Pavia, Pavia, Italy. The urban image consists of 103 spectral bands across the wavelength range from 0.43–0.86 μm. Each spectral band is composed of 610 × 340 pixels with a high spatial resolution of 1.3 m. Nine land cover classes are selected for classification in the experiments. The false color composite image and the ground reference map are shown in Fig. 3(a) and (b), respectively.

The last hyperspectral data set, Salinas, is gathered by the AVIRIS sensor over Salinas Valley, CA, USA. The spatial size is 512 × 217 pixels under a 3.7-m-high spatial resolution. A total of 204 spectral bands are used in the experiments by discarding 20 water absorption bands. There are 16 ground-truth classes as shown in the ground truth map in Fig. 5(a) and (b).

B. Experimental Setup

The optimal parameters for the proposed three methods are listed in Table I in the experiment. The range of regularization parameters λ is set as $\{1e{-}5; 1e{-}4; 1e{-}3; 1e{-}2; 1e{-}1; 1e{+}0; 1e{+}1; 1e{+}2; 1e{+}3\}$. The number of classifier T (ensemble size) is $\{10; 20; 40; 60\}$. The ratio of feature d is selected from $\{10\%; 20\%; 30\%; 40\%; 50\%; 60\%; 70\%; 80\%; 90\%\}$. The number of classes k in KNCCRT algorithms is set as: for Indian Pines, data are $\{7; 8; 9\}$; for Pavia University, data are $\{7; 8; 9\}$; and for Salinas, data are $\{8; 9; 10; 16\}$.

For Indian Pines data, 2% labeled sample of each class is picked randomly as training samples. For Pavia University data, 15 samples in each class are chosen randomly as training samples. For Salinas data, ten samples per class are selected randomly as training samples.

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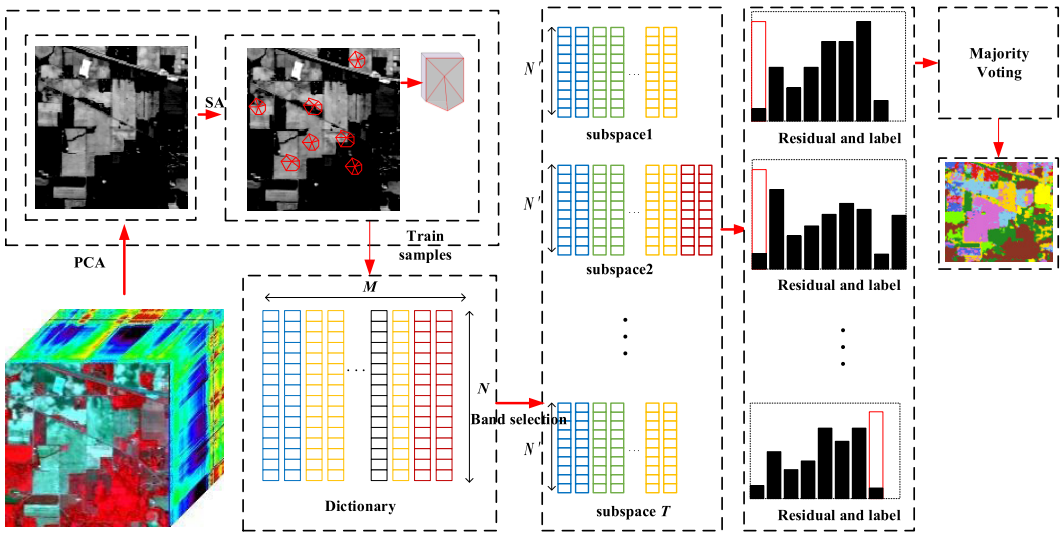


Fig. 2. Illustration of SARSKNCCRT for HSI classification.

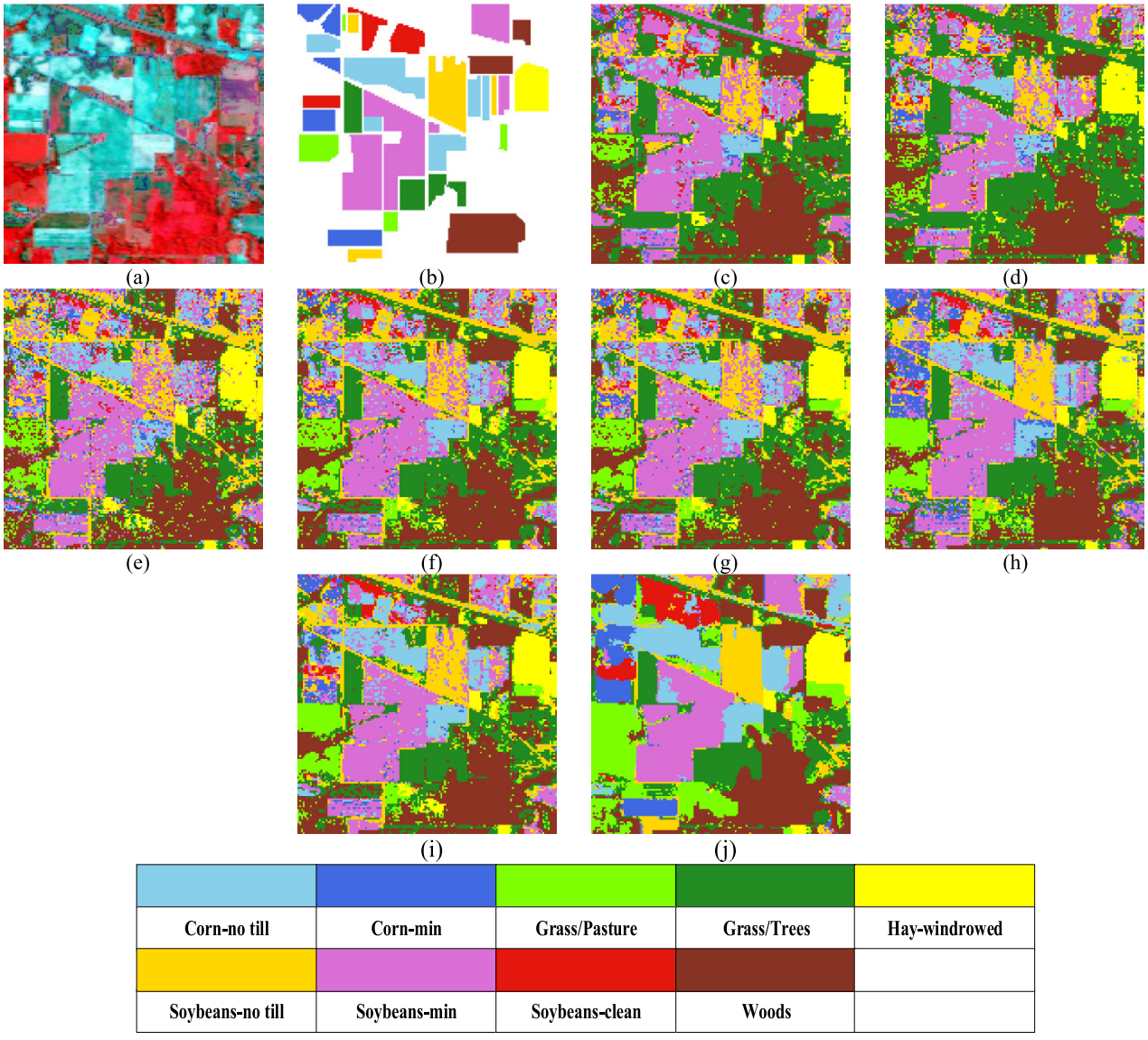


Fig. 3. Indian pines. (a) False-color image and (b) ground truth. Classification maps obtained by (c) RS, (d) RF, (e) ELM, (f) CRT, (g) KNCCRT, (h) RSELM, (i) RSKNCCRT, and (j) SARSKNCCRT.

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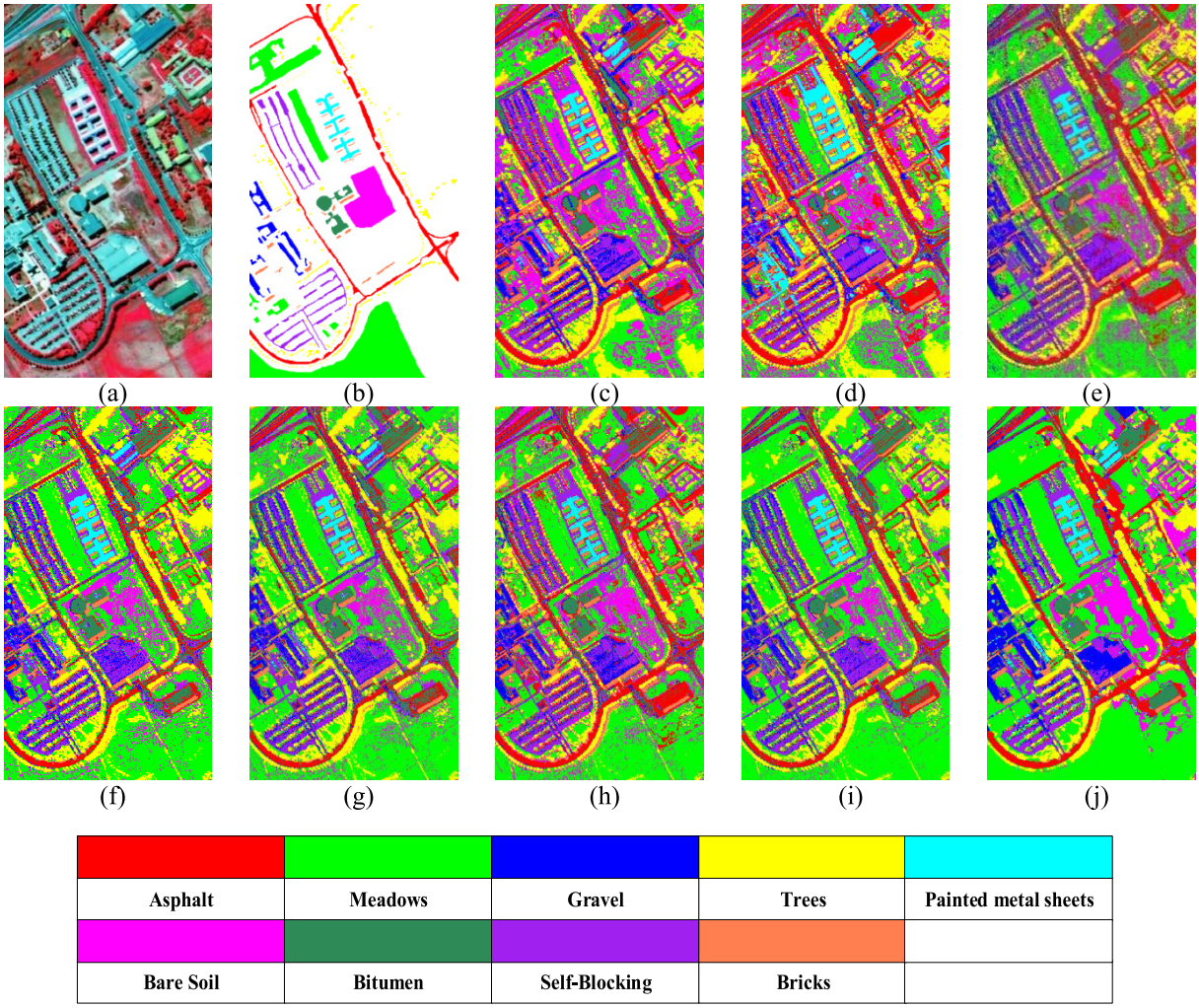


Fig. 4. Pavia University. (a) False-color image and (b) ground truth. Classification maps obtained by (c) RS, (d) RF, (e) ELM, (f) CRT, (g) KNCCRT, (h) RSELM, (i) RSKNCCRT, and (j) SARSKNCCRT.

To validate the performance of the proposed models, some classic classifiers are chosen for comparison, including RS (base classifier is DT), RF, ELM, CRT, and RSELM. Both the ELM and RSELM are operated in optimal parameters. The RS and RF classifiers are performed 20 iterations. All methods are repeated 10 times to obtain the average classification accuracy.

C. Classification Performance

To quantitatively evaluate the performance of eight algorithms, overall accuracy (OA), average accuracy (AA), kappa statistic, and individual class accuracy are used in the experiments. Ensemble times T is set as 20 for three hyperspectral data sets. The number of class k is set as 8 for three hyperspectral data sets in the experiments. For Indian Pines data set, classification performance of various methods under the optimal parameters is listed in Table II. The best results are shown in bold. The classification maps generated by the eight classifiers are shown in Fig. 3(c)–(j). As shown in Fig. 3(c)–(j), the proposed methods can produce promising classification results. Obviously, SARSKNCCRT involves the embedding of spatial information, which generate more homogeneous areas and reduce the classification noise. From Table II, the OA (%) values for eight classifiers (i.e., RS, RF, ELM, CRT, KNCCRT, RSELM, RSKNCCRT, and SARSKNCCRT) are 64.76, 63.11, 62.36, 69.25, 69.65, 76.30, 77.07, and 94.37, respectively.

Compared with RS, RF, ELM, and CRT, the proposed KNCCRT offers improvement nearly 5%, 6%, 7%, and 0.4%, respectively. It is clearly shown that the OA of RSKNCCRT increases 9% over KNCCRT, which demonstrates the effective of the RS scheme in Indian Pines data set. Furthermore, the OA value of SARSKNCCRT obtains the highest OA, AA, and kappa statistic, with the improvement of 17.30%, 14.22%, and 0.2074 over the RSKNCCRT method. The SARSKNCCRT performs the best classification results among the eight methods, with the OA nearly 95%. The OA of SARSKNCCRT is much higher than those obtained only with spectral information, demonstrating that SA can accurately model spatial information. The experimental results prove that SA regions can sufficiently exploit spatial information for Indian Pines data.

For Pavia University data set, the classification results of eight methods under the optimal parameters are shown in Table III. The land cover maps of different algorithms are shown in Fig. 4(c)–(j). As shown in Fig. 4(j), we can see

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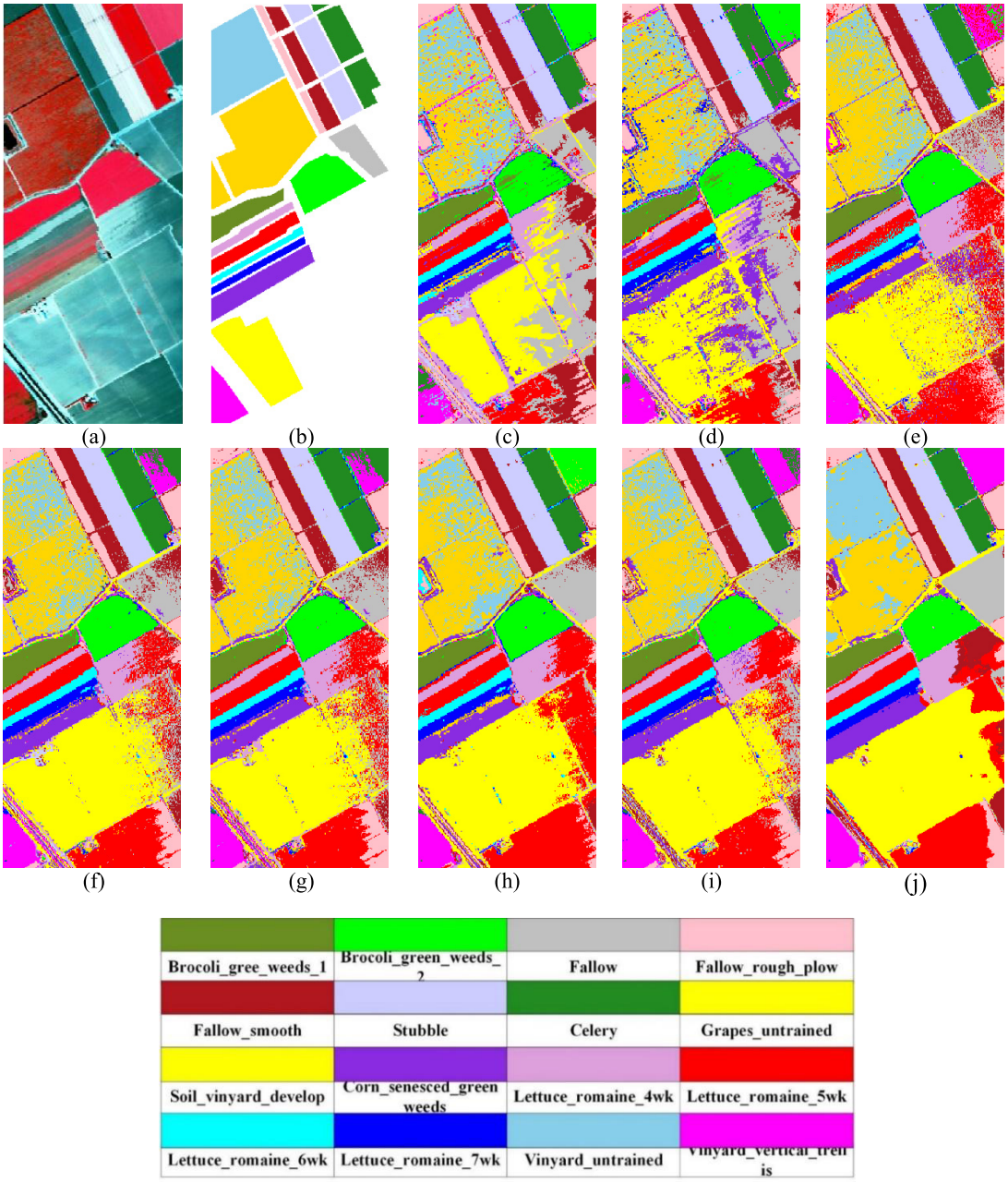


Fig. 5. Salinas. (a) False-color image and (b) ground truth. Classification maps obtained by (c) RS, (d) RF, (e) ELM, (f) CRT, (g) KNCCRT, (h) RSELM, (i) RSKNCCRT, and (j) SARSKNCCRT.

that the best classification accuracy comes from SRSKNCCRT algorithm.

SARSKNCCRT method based on spectral–spatial technique results in classification maps with more homogeneous region when compared with the classification result using spectral information. From Table III, the OA (%) values for eight classifiers (i.e., RS, RF, ELM, CRT, KNCCRT, RSELM, RSKNCCRT, and SARSKNCCCRT) are 63.48, 63.62, 68.11, 76.23, 76.40, 78.28, 79.08, and 90.96, respectively. Similarly, KNCCRT algorithm is slightly better than CRT algorithm. Due to RS ensemble strategy, RSKNCCRT achieves better performance than the KNCCRT- and ELM-based ensemble methods. Specifically, RSKNCCRT achieves 2.6% OA higher than KNCCRT, 1.8% higher than RSELM, and 16% higher than RF, respectively. However, SARSKNCCRT algorithm can achieve the highest OA over 90%. It is suggested that the classification results with SA constraint significantly outperformed those only considering spectral information.

For the Salinas data set, the eight visual classification maps are shown in Fig. 5(c)–(j). As shown in Table IV, the OA (%) values for eight classifiers (i.e., RS, RF, ELM, CRT, KNCCRT, RSELM, RSKNCCRT, and SARSKNCCCRT) are 80.51, 80.51, 82.86, 84.94, 85.12, 85.60, 86.36, and 95.94, respectively. Similarly, KNCCRT generates more accurate results than CRT. The improvement of RSKNCCRT over KNCCRT demonstrates that the RS can alleviate the curse of high dimension. And ELM-based ensemble classifiers yield better performance than the individual ELM classifier. For

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TABLE II
CLASSIFICATION ACCURACY (IN PERCENT) FOR INDIAN PINES

Class	Train	Test	RS	RF	ELM	CRT	KNCCRT	RSELM	RSKNC CRT	SARS KNCCRT
C1	29	1399	42.43	46.79	53.27	61.18	61.58	69.73	72.16	**90.37**
C2	17	813	25.29	28.89	39.41	52.78	52.97	73.38	69.58	**93.04**
C3	10	473	50.02	46.38	77.70	90.43	90.35	93.70	94.57	**97.13**
C4	15	715	95.94	94.43	85.96	89.10	89.25	91.37	88.17	**97.29**
C5	10	468	99.66	98.16	91.48	94.31	94.44	99.91	98.29	**100**
C6	20	952	47.5	40.42	45.02	53.52	53.39	69.89	69.31	**88.79**
C7	50	2405	78.98	74.98	55.21	61.13	61.87	65.59	66.87	**95.58**
C8	12	581	24.48	22.93	33.94	45.83	45.79	77.22	66.39	**90.19**
C9	26	1239	94.85	92.28	89.03	96.21	96.43	99.15	98.83	**99.71**
Overall Accuracy (%)			64.76	63.11	62.36	69.25	69.65	76.30	77.07	**94.37**
Average Accuracy (%)			62.13	60.58	63.45	71.60	71.78	82.22	80.46	**94.68**
Kappa			0.5763	0.5585	0.5485	0.6363	0.6405	0.7268	0.7273	**0.9347**

TABLE III
CLASSIFICATION ACCURACY (IN PERCENT) FOR PAVIA UNIVERSITY

Class	Train	Test	RS	RF	ELM	CRT	KNCCRT	RSELM	RSKNC CRT	SARS KNCCRT
1	15	6616	62.52	68.35	80.42	95.49	95.87	88.83	96.82	**98.67**
2	15	18634	54.20	51.25	88.89	89.89	90.36	91.11	90.66	**95.61**
3	15	2084	61.77	57.84	40.69	50.29	51.02	52.74	52.37	**77.54**
4	15	3049	93.54	95.31	66.12	68.35	68.31	76.33	71.41	**98.98**
5	15	1330	99.16	98.92	99.95	96.89	98.25	**99.98**	99.06	96.50
6	15	5014	61.17	63.17	45.38	60.65	61.40	62.71	68.90	**90.52**
7	15	1315	**91.47**	85.35	33.50	39.66	38.60	43.52	41.59	57.75
8	15	3667	59.37	65.71	60.35	66.95	70.46	74.26	71.58	**85.31**
9	15	932	99.79	99.96	99.96	100	100	99.98	100	**100**
Overall Accuracy (%)			63.48	63.62	68.11	76.23	76.40	78.28	79.08	**90.96**
Average Accuracy (%)			75.89	76.21	68.36	74.24	74.92	76.61	76.93	**88.99**
Kappa			0.5530	0.5571	0.5999	0.6942	0.6980	0.7253	0.7687	**0.8855**

the effect of local SA region, SARSKNCCRT shows the best results, which are significantly better than RSKNCCRT, especially for classes 3, 8, and 15.

V. DISCUSSION

A. Parameter Analysis for k

The impact of different classes number k is explored. The number of classes k has been set as shown in Section IV-B; other parameters have been fixed, as shown in Table I. For Indian Pines and Pavia University data sets, KNCCRT algorithm represents CRT algorithm actually when $k = 9$. As shown in Fig. 6(a), KNCCRT shows better classification results than CRT in Indian Pines data set. The value of k is set as {7, 8, 9}, the OA increases firstly and then decreases. And KNCCRT offers the best classification performance when $k = 8$. The accuracy of KNCCRT ($k = 8$) increases with the growing regularization parameters λ and reaches maximum when $\lambda = 1e - 1$, and then the accuracy becomes decrease when λ is greater than $1e - 1$.

Similarly, from Fig. 6(b), the OA increases first and then decreases with the value of k. It can be seen that the KNCCRT achieves the best results when k is equal to 8 in Pavia University data set. The most suitable λ set in KNCCRT is $1e-1$ for Pavia University data. Table V shows the execution time of KNCCRT for three experimental data. With decreases of k, less time is consumed as expected. To balance the classification accuracy and time consumption, the number of k is adopted as 8 for Indian Pines data and Pavia University. The experimental results demonstrate that KNCCRT improves the accuracy and time efficiency simultaneously.

For Salinas data set, k is set as {8, 9, 10, 16}. And KNCCRT algorithm represents CRT algorithm when $k = 16$ in Salinas data. From Fig. 6(c), the OA decreases as the value of k increases. It is obvious that KNCCRT shows superior performance than CRT, especially when k is adopted as 8. So, the number of k is employed as 8 for KNCCRT algorithm. And the most suitable λ set is $1e-2$ in KNCCRT for Salinas data set.

B. Parameter Analysis for T

The impact of ensemble size T is investigated in this article. The number of classes T has been set as shown in Section IV-B, and other parameters are fixed, as shown

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TABLE IV
CLASSIFICATION ACCURACY (IN PERCENT) FOR SALINAS

Class	Train	Test	RS	RF	ELM	CRT	KNCCRT	RSELM	RSKNC CRT	SARS KNCCRT
1	10	1999	96.23	96.20	99.69	99.93	99.91	99.97	99.94	**100**
2	10	3716	95.11	98.01	99.25	99.11	99.16	99.28	99.17	**99.95**
3	10	1966	83.16	85.42	84.12	92.12	92.93	96.53	93.71	**97.81**
4	10	1384	98.97	98.37	98.92	96.69	98.27	98.53	98.43	**97.60**
5	10	2668	92.22	92.40	84.35	94.37	94.63	93.18	97.60	**99.14**
6	10	3949	98.18	97.85	99.95	99.88	99.99	99.99	99.98	**100**
7	10	3569	98.77	97.69	99.30	98.94	98.53	99.40	98.99	**99.92**
8	10	11261	58.19	59.18	70.65	73.66	74.11	72.91	72.22	**91.56**
9	10	6193	93.85	95.07	97.98	98.29	98.25	98.57	97.77	**98.52**
10	10	3268	71.54	76.18	86.57	90.50	89.59	90.02	91.66	**97.11**
11	10	1058	81.34	81.76	84.90	87.75	86.36	90.11	91.38	**95.00**
12	10	1917	95.08	96.63	88.67	96.18	95.35	92.91	94.15	**95.04**
13	10	906	97.77	94.94	79.67	92.77	92.55	88.62	93.96	**99.12**
14	10	1060	91.38	93.17	91.25	89.78	90.16	**95.03**	89.81	93.61
15	10	7258	61.27	53.80	51.20	54.31	55.61	53.53	56.66	**84.29**
16	10	1797	79.91	84.56	99.66	99.23	99.40	99.64	**99.77**	99.72
Overall Accuracy (%)			80.51	80.51	82.86	84.97	85.12	85.66	86.36	**93.57**
Average Accuracy (%)			87.06	87.58	88.51	91.47	91.55	91.76	92.20	**95.94**
Kappa			0.7840	0.7840	0.8099	0.8398	0.8426	0.8366	0.8473	**0.9294**

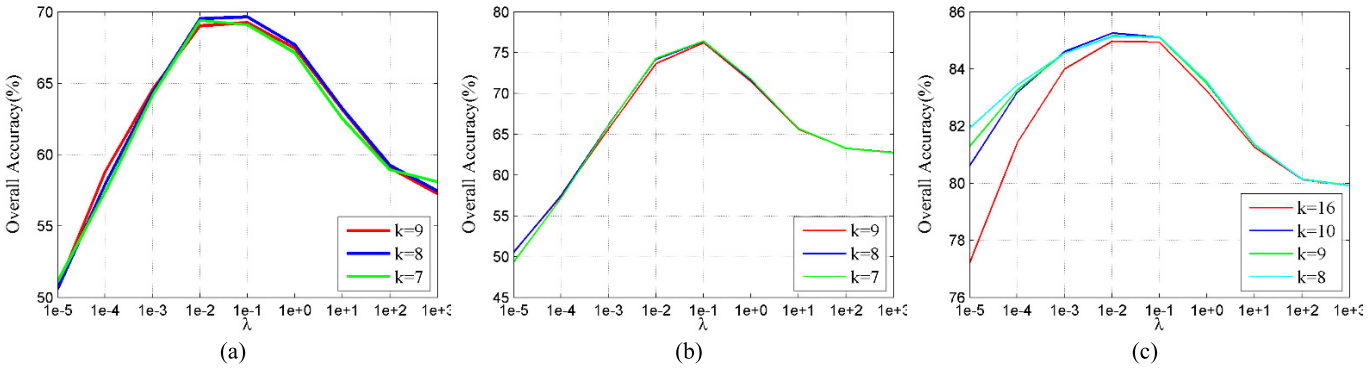


Fig. 6. Overall accuracy of different algorithms for three data sets with varying k. (a) Indian Pines. (b) Pavia University. (c) Salinas.

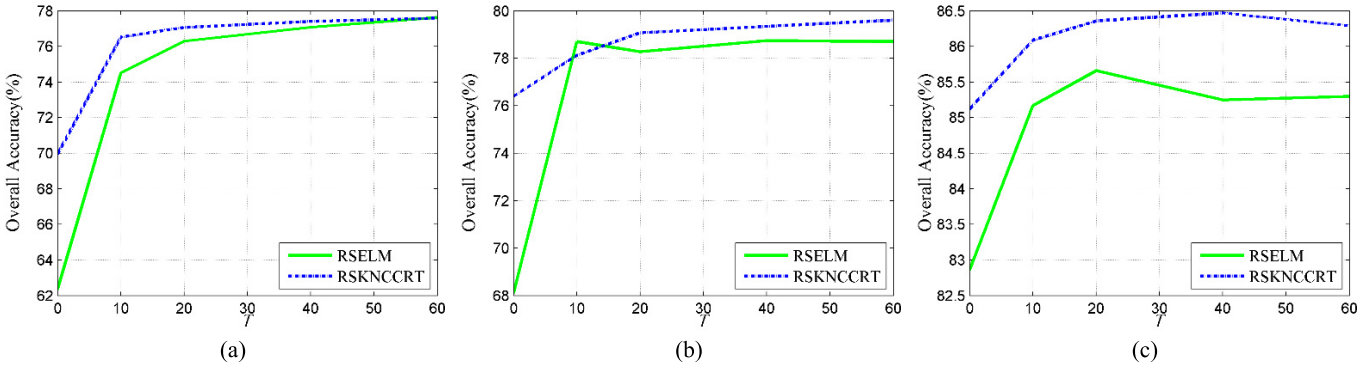


Fig. 7. Overall accuracy of different algorithms for three data sets with varying T. (a) Indian Pines. (b) Pavia University. (c) Salinas.

in Table I. The zero value of the x-coordinate represents KNCCRT classifier. From Fig. 7(a)–(c), it is obvious that the OA was significantly improved by growing the ensemble size between 10 and 20 for three hyperspectral data sets. However, ensemble times continue to increase, and the OA does not increase further for both RSELM and RSKNCCRT.

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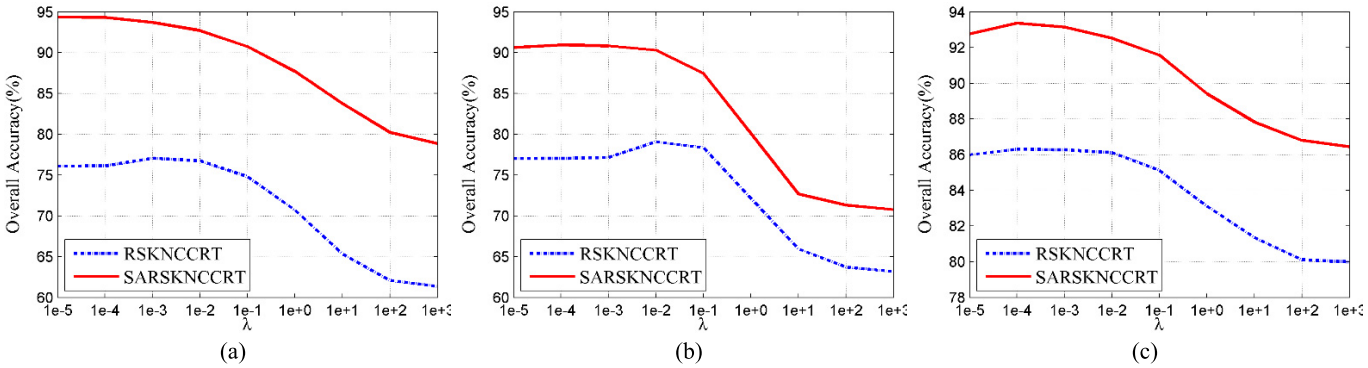


Fig. 8. Overall accuracy of different algorithms for three data sets with varying λ. (a) Indian Pines. (b) Pavia University. (c) Salinas.

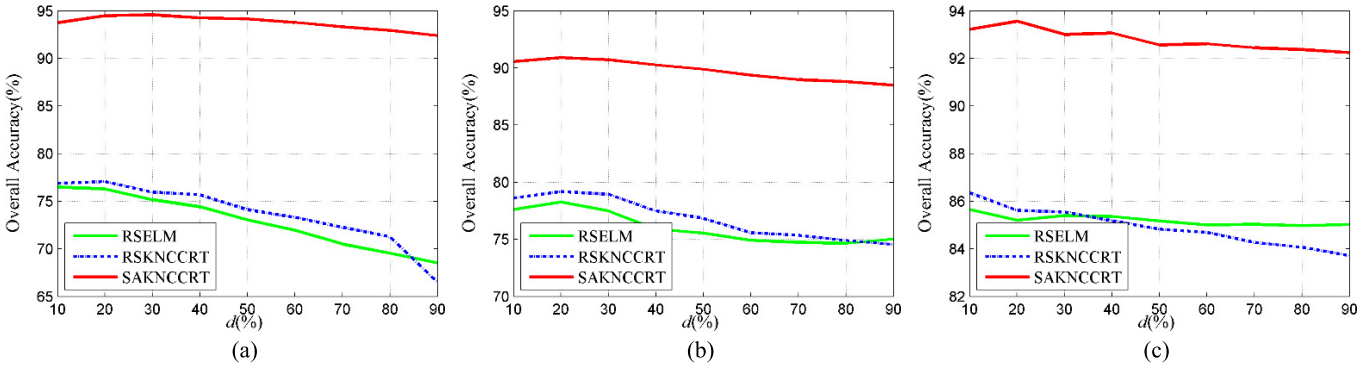


Fig. 9. Overall accuracy of different algorithms for three data sets with varying d. (a) Indian Pines. (b) Pavia University. (c) Salinas.

TABLE V
EXECUTION TIME (S) FOR EXPERIMENTAL DATA SETS

Data		Indian Pines	Pavia University	Salinas
KNCCRT	k=7	0.94(s)	6.76(s)	100.22(s)
	k=8	1.10(s)	9.65(s)	103.47(s)
	k=9	1.37(s)	10.10(s)	124.00(s)
	k=10			124.39(s)
	k=16			175.15(s)

With the number of base classifiers increases, the diversity between base classifiers decreases instead, which results in poor classification accuracy. Although a larger ensemble size sometimes provides enhanced classification accuracy, it results in higher computational complexity naturally. In practical applications, it is needed to consider the computational complexity and classification performance comprehensively. Thus, KNCCRT ensemble approaches obtain satisfying accuracy with 20 iterations, and additional iterations cannot obviously increase the accuracy. Therefore, the number of T is adopted as 20 in Tables II–IV. From Fig. 7(a)–(c), it is clearly shown that RSKNCCRT outperforms RSELM for three hyperspectral data sets.

C. Parameter Analysis for λ

The sensitiveness of the three proposed methods with varying regularization parameter λ are evaluated in the experiments. The range of λ has been set as in Section IV-B, and other parameters have been fixed, as shown in Table I. As shown in Fig. 8(a)–(c), it illustrates the OA tendency of KNCCRT, RSKNCCRT, and SARSKNCCRT with different regularization parameter λ for three hyperspectral data sets. It is observed that the OA of the RSKNCCRT and SARSKNCCRT is improved with the growing regularization parameter and then would generally decrease after reaching the maximum. In Fig. 8, RSKNCCRT provides the best classification performance for three hyperspectral data sets when $\lambda = 1e-3$, $\lambda = 1e-2$, and $\lambda = 1e-4$, respectively. For SARSKNCCRT, the most suitable λ for three hyperspectral data sets is $1e-5$, $1e-4$, and $1e-4$, respectively. Compared with the RSKNCCRT, it is noted that incorporation of SA neighborhood information in the proposed SARSKNCCRT algorithms leads to a significant improvement.

D. Parameter Analysis for Feature Numbers

The number of features in a subspace is also the key parameter to investigate. To find an optimal number of features, we analyze the sensitivity of different subspace sizes for three hyperspectral data sets. The number of features d has been set as in Section IV-B, and other parameters are fixed, as shown in Table I. As shown in Fig. 9(a) and (b), with the increase of feature size, the overall classification accuracy of RSELM, RSKNCCRT, and SARSKNCCRT algorithms increases, and then the OA maintains as a certain value or even slightly decreases. For RSKNCCRT, when the feature size is greater

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TABLE VI
CLASSIFICATION ACCURACIES OBTAINED FROM THE PROPOSED METHODS IN COMPARISON WITH OTHER SPATIAL–SPECTRAL CLASSIFIERS

Data	Indian Pines				Pavia University				Salinas			
Method	JCR	WJCR	RSELM EMAP	SARS KNCCRT	JCR	WJCR	RSELM EMAP	SARS KNCCRT	JCR	WJCR	RSELM EMAP	SARS KNCCRT
OA(%)	73.47	74.26	88.84	**94.37**	83.88	86.10	90.78	**90.96**	86.55	86.57	92.31	**93.57**
AA(%)	78.97	79.89	89.40	**94.68**	82.49	86.04	89.56	**88.99**	91.43	90.35	95.49	**95.94**
Kappa	0.6813	0.7141	0.8688	**0.9347**	0.7901	0.8179	0.8802	**0.8855**	0.8506	0.8510	0.9145	**0.9294**

TABLE VII
RUNNING TIMES (IN SECONDS) FOR DIFFERENT METHODS

Running Time	RS	RF	ELM	CRT	KNCCRT	RSELM	RSKNC CRT	SARS KNCCRT
Indian Pines	4.26	0.15	0.14	1.37	1.10	2.02	87.64	283.67
Pavia University	5.86	4.71	0.30	10.10	9.65	5.53	50.13	105.89
Salinas	20.18	0.28	0.24	175.15	103.47	3.68	750.53	903.31

than 80% of the original feature set, the classification accuracy is the worst of all, even drops to 69% in the Indian Pines data set. Difference between the maximum and minimum accuracies is more than 9% in RSKNCCRT method. From Fig. 9(c), the maximum of RSKNCCRT and RSELM is reached when the number of features is 10% in Salinas data. Then, with the increase of number of features, the OA of RSKNCCRT and RSELM is reduced. In general, for RSKNCCRT ensemble, it is recommended to adopt the number of 10%–30% of features for classification in the experiments. Compared with RSELM, RSKNCCRT yields the superior classification performance in three hyperspectral data sets. The SARSKNCCRT algorithm produces the best classification results among all the classifiers for three data sets.

E. Comparison With Spatial–Spectral Methods

Table VI presents the classification results obtained from the proposed methods SARSKNCCRT in comparison with other spatial–spectral classifiers. There are three spectral–spatial classifiers, such as JCR [34], WJCR [35], and RSELM-EMAP [42]. JCR and WJCR methods incorporate window-based spatial features. Its gain great accuracy improvement compared with CR-based using spectral information. RSELM-EMAP shows better performance than JCR, WJCR, and RSELM. When Extended Morphological Attribute Profiles (EMAPs) feature is treated as input, classification results of these classes are greatly improved. The proposed method SARSKNCCRT yields the highest classification results for three hyperspectral data sets, suggesting the effectiveness of the SA constraint.

F. Complexity Analysis and Computing Time

From the aspects of complexity analysis, the complexity of KNCCRT and SARSKNCCRT is very high than others. KNCCRT needs to obtain the k-nearest class by calculating the distance between testing sample and subdictionary; and SARSKNCCRT needs to obtain both k-nearest class and SA region of target sample. All the algorithms are implemented using MATLAB R2014a environment on a 2.60-GHz CPU with 8 GB of RAM. The computing time of these algorithms is listed in Table VII. KNCCRT takes less time than CRT owing to reduced dictionary. It obviously observed that ELM-related classifiers are very fast. The reason is that it randomly initializes weights. The computation time of RSELM is extremely low compared with other ensemble algorithms. RSKNCCRT requires longer computational time than KNCCRT. In order to calculate the SA neighborhood of testing and training samples, the proposed SARSKNCCRT costs much more execution time than RSKNCCRT in the experiments.

VI. CONCLUSION

In this article, in order to improve the performance of CRT algorithms, RSKNCCRT is designed to select the training samples from k-nearest class to represent a testing sample. RSKNCCRT ensembles have been proposed by integrating KNCCRT and RS to offer better classification performance. Then, SA regions for testing and training sample are constructed adaptively for incorporating spatial information effectively. The experimental results proved that KNCCRT algorithm improves the classification accuracy and computational efficiency compared with CRT algorithm. RS strategy can provide superior performance. Spectral–spatial classification approaches computed by SA region achieve the best performance for three data sets.

However, the number of bands in a feature subset needs to be predefined in the experiments. Therefore, we plan to design an effective strategy to adaptively determine the optimal number of bands in a feature subset for RS-based ensemble in the future.

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